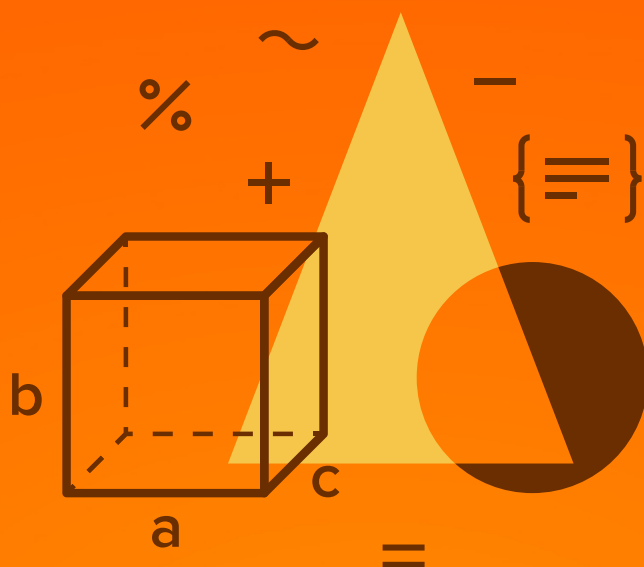




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Function & Inverse Trigonometric Functions

English Medium

EXERCISE (O-1)

Straight Objective Type

1. Let $P(x) = kx^3 + 2k^2x^2 + k^3$. The sum of all real numbers k for which $(x-2)$ is a factor of $P(x)$, is
(A) 4 (B) 8 (C) -4 (D) -8
FN0053
2. Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ and $P(6) = 6$, then find the value of $P(7)$.
(A) 700 (B) 727 (C) 650 (D) 725
FN0158
3. The domain of definition of the function, $y(x)$ is given by the equation, $2^x + 2^y = 2$ is - **FN0199**
[JEE 2000 (Scr.), 1M]
(A) $0 < x \leq 1$ (B) $0 \leq x \leq 1$ (C) $-\infty < x \leq 0$ (D) $-\infty < x < 1$
4. Range of function $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$ is given by
(A) $(0, \infty)$ (B) $\left[\frac{1}{2}, 1 \right]$ (C) $[1, 2]$ (D) $\left[\frac{1}{4}, 1 \right]$
FN0064
5. Range of $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}$; $x \in \left(0, \frac{\pi}{2} \right)$ is -
(A) $(0, 1)$ (B) $(1, \infty)$ (C) $(-1, 0)$ (D) $(-\infty, -1)$
FN0065
6. If $f(x, y) = \max(x, y) + \min(x, y)$ and $g(x, y) = \max(x, y) - \min(x, y)$, then the value of $f\left(g\left(-\frac{2}{3}, -\frac{3}{2}\right), g(-3, -4)\right)$ is greater than -
(A) 1 (B) 2 (C) 3 (D) 4
FN0066
7. The number of integral values of x satisfying the inequality $[x-5][x-3] + 2 < [x-5] + 2[x-3]$ (where $[.]$ represents greatest integer function) is -
(A) 0 (B) 1 (C) 2 (D) 3
FN0067
8. Suppose, $f(x, n) = \sum_{k=1}^n \log_x \left(\frac{k}{x} \right)$, then the value of x satisfying the equation $f(x, 10) = f(x, 11)$, is
(A) 9 (B) 10 (C) 11 (D) none
FN0068

9. Let $f(x) = \frac{x}{1-x}$ and let α be a real number. If $x_0 = \alpha$, $x_1 = f(x_0)$, $x_2 = f(x_1)$, & $x_{2011} = -\frac{1}{2012}$ then the value of α is -

(A) $\frac{2011}{2012}$ (B) 1 (C) 2011 (D) -1

FN0070

10. If $f_1(x) = 2^{f_2(x)}$, where $f_2(x) = 2012^{f_3(x)}$, where $f_3(x) = \left(\frac{1}{2013}\right)^{f_4(x)}$, where $f_4(x) = \log_{2013} \log_x 2012$, then the range of $f_1(x)$ is -

(A) $(2, \infty)$ (B) $(2012, \infty)$ (C) $(0, \infty)$ (D) $(-\infty, \infty)$

FN0071

11. Let f be a function satisfying $f(xy) = \frac{f(x)}{y}$ for all positive real numbers x and y . If $f(30) = 20$, then the value of $f(40)$ is-

(A) 15 (B) 20 (C) 40 (D) 60

FN0200

12. Let $f : \mathbb{R} - \left\{\frac{-15}{2}\right\} \rightarrow \mathbb{R} - \left\{\frac{1}{2}\right\}$ be defined by $f(x) = \frac{x+10}{2x+15}$ then $f(x)$ is-

(A) one-one but not onto (B) many one but not-onto
(C) one-one and onto (D) many one and onto

FN0046

13. $f : \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = \frac{2x^2 - 5x + 3}{8x^2 + 9x + 11}$, then f is -

(A) one-one onto (B) many-one onto (C) one-one into (D) many one into

FN0047

14. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} x+3 & , x \in \text{rational} \\ 4x & , x \in \text{irrational} \end{cases}$,

$$g(x) = \begin{cases} x + \sqrt{5} & , x \in \text{irrational} \\ -x & , x \in \text{rational} \end{cases} \text{ then } (f - g)(x) \text{ is-}$$

(A) one-one & onto (B) neither one-one nor onto
(C) one-one but not onto (D) onto but not one-one

FN0072

15. Let $f : A \rightarrow B$ be an onto function such that $f(x) = \sqrt{x-2-2\sqrt{x-3}} - \sqrt{x-2+2\sqrt{x-3}}$, then set 'B' is-

(A) $[-2, 0]$ (B) $[0, 2]$ (C) $[-3, 0]$ (D) $[-1, 0]$

FN0073

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^3 + ax^2 + bx - 8$. If $f(x) = 0$ has three real roots & $f(x)$ is a bijective function, then $(a + b)$ is equal to
 (A) 0 (B) 6 (C) -6 (D) 12

FN0074

17. Let $f : X \rightarrow Y$ be a function such that $f(x) = \sqrt{x-2} + \sqrt{4-x}$, then the set of X and Y for which $f(x)$ is both injective as well as surjective, is -

FN0201

- (A) $[2,4]$ and $[\sqrt{2},2]$ (B) $[3,4]$ and $[\sqrt{2},2]$
 (C) $[2,4]$ and $[1,2]$ (D) $[2,3]$ and $[1,2]$

18. If $f : \mathbb{R} \rightarrow \mathbb{R}$ & $f(x) = \frac{\sin([x]\pi)}{x^2 + 2x + 3} + 2x - 1 + \sqrt{x(x-1)} + \frac{1}{4}$ (where $[x]$ denotes integral part of x), then $f(x)$ is -

- (A) one-one but not onto (B) one-one & onto
 (C) onto but not one-one (D) neither one-one nor onto

FN0048

19. If $f(x) = px + q$ and $f(f(f(x))) = 8x + 21$, where p and q are real numbers, then $p + q$ equals-

- (A) 3 (B) 5 (C) 7 (D) 11

FN0202

20. $f(x)$ and $g(x)$ are linear function such that for all x , $f(g(x))$ and $g(f(x))$ are Identity functions. If $f(0) = 4$ and $g(5) = 17$, compute $f(2006)$.

- (A) 122 (B) 102 (C) 96 (D) 205

FN0168

21. $f(x) = [x-1] + \{x\}^{[x]}$, $x \in (1,3)$, then $f^{-1}(x)$ is -

(where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function)

- (A) $\begin{cases} x+1 & x \in (1,2) \\ 2+\sqrt{x-1} & x \in [2,3) \end{cases}$ (B) $\begin{cases} x-1 & x \in (1,2) \\ 2-\sqrt{x-1} & x \in [2,3) \end{cases}$
 (C) $\begin{cases} x-1 & x \in (0,1) \\ 2-\sqrt{x-1} & x \in [1,2) \end{cases}$ (D) $\begin{cases} x+1 & x \in (0,1) \\ 2+\sqrt{x-1} & x \in [1,2) \end{cases}$

FN0076

22. Let $f(x) = \frac{2x-1}{x+3}$. If $f^{-1} = \frac{ax+b}{c-x}$, then $a + b + c$ is

- (A) 7 (B) 6 (C) 5 (D) 4

FN0137

23. Let $f : (0, \infty) \rightarrow (1, \infty)$ be a function such that $f(x) = 1 + \frac{3}{2}\sqrt{x}$ and $g(x)$ is inverse of $f(x)$ then the point where $f(x)$ and $g(x)$ intersect is

FN0203

- (A) $\left(\frac{1}{4}, \frac{1}{4}\right)$ (B) (4, 2) (C) (4, 4) (D) Does not exist

24. If the function $f : \left[\frac{7}{2}, \infty\right) \rightarrow \left[-\frac{9}{4}, \infty\right)$ be defined by $f(x) = x^2 - 7x + 10$ then $f^{-1}(x)$ is- **FN0204**

(A) $\frac{7 + \sqrt{9 + 4x}}{2}$ (B) $\frac{7 \pm \sqrt{9 + 4x}}{2}$
 (C) $\frac{7 + \sqrt{9 - 4x}}{2}$ (D) $\frac{7 \pm \sqrt{9 - 4x}}{2}$

25. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \ln(x + \sqrt{x^2 + 1})$, then number of solutions of $|f^{-1}(x)| = e^{-|x|}$ is :-
 (A) 1 (B) 2 (C) 3 (D) Infinite

FN0075

26. Which of the following functions is an odd function ?

(A) $|x - 2| + (x + 2) \operatorname{sgn}(x + 2)$ (B) $\frac{1}{x(e^x - 1)} + \frac{1}{2x}$
 (C) $\log(\sin x + \sqrt{1 + \sin^2 x})$ (D) $e^{-4x}(e^{2x} - 1)^4$
 (where $\operatorname{sgn}(x)$ denotes signum function of x)

FN0077

27. Let $f(x) = 2x - \left\{\frac{x}{\pi}\right\}$ and $g(x) = \cos x$, where $\{.\}$ denotes fractional part function, then period of $g \circ f(x)$ is -

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{2}$ (D) $\frac{\pi}{4}$

FN0079

28. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function such that $f(10 + x) = f(10 - x) \forall x \in \mathbb{R}$ and $f(20 + x) = -f(20 - x) \forall x \in \mathbb{R}$. Then which of the following statements is true -
 (A) $f(x)$ is odd and periodic (B) $f(x)$ is odd and aperiodic
 (C) $f(x)$ is even and periodic (D) $f(x)$ is even and aperiodic

FN0081

29. Period of $f(x) = \{x\} + \left\{x + \frac{1}{3}\right\} + \left\{x + \frac{2}{3}\right\}$ is equal to (where $\{.\}$ denotes fractional part function)

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

FN0078

30. Period of function $f(x) = \min\{\sin x, |x|\} + \frac{x}{\pi} - \left[\frac{x}{\pi}\right]$ (where $[.]$ denotes greatest integer function) is -
 (A) $\pi/2$ (B) π (C) 2π (D) 4π

FN0054

EXERCISE (O-2)

Multiple Correct Answer Type

- 1.** Which of the following function(s) have the same domain and range?

(A) $f(x) = \sqrt{1 - x^2}$

(B) $g(x) = \frac{1}{x}$

(C) $h(x) = \sqrt{x}$

(D) $l(x) = \sqrt{4 - x}$

FN0055

2. Which of the following pair(s) of function have same graphs?

$$(A) f(x) = \frac{\sec x}{\cos x} - \frac{\tan x}{\cot x}, g(x) = \frac{\cos x}{\sec x} + \frac{\sin x}{\operatorname{cosec} x}$$

(B) $f(x) = \text{sgn}(x^2 - 4x + 5)$, $g(x) = \text{sgn}\left(\cos^2 x + \sin^2\left(x + \frac{\pi}{3}\right)\right)$ where sgn denotes signum function.

(C) $f(x) = e^{\ln(x^2+3x+3)}$, $g(x) = x^2 + 3x + 3$

$$(D) f(x) = \frac{\sin x}{\sec x} + \frac{\cos x}{\operatorname{cosec} x}, g(x) = \frac{2\cos^2 x}{\cot x}$$

FN0056

3. Let $f(x) = \begin{cases} x^2 - 3x + 4 & ; \ x < 3 \\ x + 7 & ; \ x \geq 3 \end{cases}$ and $g(x) = \begin{cases} x + 6 & ; \ x < 4 \\ x^2 + x + 2 & ; \ x \geq 4 \end{cases}$, then which of the following is/are true -

(A) $(f + g)(1) = 9$

(B) $(f - g)(3.5) = 1$

$$(C) (f \cdot g)(0) = 24$$

$$(D) \left(\frac{f}{g} \right) (5) = \frac{8}{3}$$

FN0058

4. Let $f(x) = x^2 + 3x + 2$, then number of solutions to -

(A) $f(|\mathbf{x}|) = 2$ is 1

(B) $f(|x|) = 2$ is 3

(C) $|f(x)| = 0.125$ is 4

(D) $|f(|x|)| = 0.125$ is 8

FN0060

5. For each real x , let $f(x) = \max\{x, x^2, x^3, x^4\}$, then $f(x)$ is -

(A) x^4 for $x \leq -1$

(B) x^2 for $-1 < x \leq 0$

$$(C) \quad f\left(\frac{1}{2}\right) = \frac{1}{2}$$

(D) $f\left(\frac{1}{2}\right) = \frac{1}{4}$

FN0061

6. For the function $f(x) = |x + 3| - |x + 1| - |x - 1| + |x - 3|$, identify correct option(s)

(A) Range of $f(x)$ is $(-\infty, 4]$

(B) maximum value of $f(x)$ is 4

(C) $f(x) = 4$ has infinite solutions

(D) $f(x) = 0$ has infinite solutions

FN0083

7. The values of x in $[-2\pi, 2\pi]$, for which the graph of the function $y = \sqrt{\frac{1+\sin x}{1-\sin x}} - \sec x$ and

$$y = -\sqrt{\frac{1-\sin x}{1+\sin x}} + \sec x, \text{ coincide are}$$

- (A) $\left[-2\pi, -\frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right]$ (B) $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
 (C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) $[-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$

FN0084

8. Which of the following statement(s) is(are) correct ?

- (A) If f is a one-one mapping from set A to A , then f is onto.
 (B) If f is an onto mapping from set A to A , then f is one-one
 (C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ f$ is injective, then f must be injective.
 (D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

FN0085

9. Consider a function $f(x) = \frac{|x+1|}{x+1} + \frac{x-1}{|x-1|}$, then choose correct option(s) regarding function f

- (A) many one (B) odd function
 (C) bounded function (D) range contain exactly three integers

FN0205

10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = 2x + 1$, $h: \mathbb{R} \rightarrow A$, $h(x) = f(x) + g(x)$, then- FN0206

- (A) $f(x)$ is bijective function (B) $g(x)$ is bijective function
 (C) if $h(x)$ is bijective function then A is $[0, \infty)$ (D) if $h(x)$ is onto function then A is $[0, \infty)$

11. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. Number of

possible distinct intersection(s) of these graph is

- (A) 0 (B) 2 (C) 3 (D) 4

FN0086

12. Let $f(x) = \begin{cases} x^2 & ; 0 < x < 2 \\ 2x-3 & ; 2 \leq x < 3 \\ x+2 & ; x \geq 3 \end{cases}$ Then :-

- (A) $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\} = f\left(\frac{3}{2}\right)$ (B) $1 + f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\} = f\left(\frac{5}{2}\right)$
 (C) $f\{f(1)\} = f(1) = 1$ (D) none of these

FN0087

13. If $\{x\} = \frac{2}{3}$ & $\left[x + \left\{ x + \left[x + \{x + \dots\dots 100 \text{ times} \} \right] \right\} \right] = 5$, then -

- (A) $x = \frac{14}{3}$ (B) $[x] = 5$ (C) $x = \frac{17}{3}$ (D) $[x] = 4$

(where $[.]$ & $\{.\}$ denotes greatest integer function & fractional part function respectively)

FN0088

14. Given $f(1) = 2$ and $f(n+1) = \frac{f(n)-1}{f(n)+1} \forall n \in \mathbb{N}$, then-

- (A) $f(2015) = -\frac{1}{2}$ (B) $(f(2012))^{f(2013)} = 9$ (C) $f(1001) = 2$ (D) $f(2015) = -3$

FN0089

15. Which of the following is/are true?

- (A) $f(x) = e^x$ and $g(x) = \ln x$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
 (B) $f(x) = \frac{2+x}{3-x}$ and $g(x) = \frac{(3x-2)}{(x+1)}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
 (C) $f(x) = 4x + 7$ and $g(x) = \frac{(x-7)}{4}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
 (D) $f(x) = x^3 + 1$ and $g(x) = (x-1)^{1/3}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)

FN0090

16. If $g(x) = x^2 - x + 1$ and $f(x) = \sqrt{\frac{1}{x} - x}$, then -

- (A) Domain of $f(g(x))$ is $[0,1]$ (B) Range of $f(g(x))$ is $\left[0, \frac{7}{2\sqrt{3}}\right]$
 (C) $f(g(x))$ is many-one function (D) $f(g(x))$ is unbounded function

FN0091

17. Let $f: \{1,2,3,4,5\} \rightarrow \{1,2,3,4,5\}$ is such that $f(x)$ is a one-one function satisfying following condition $f(x) = x + 1$ if and only if x is even (i.e. $f(3) \neq 4$, $f(4) = 5$ etc). Then $f^{-1}(2)$ can be-

- (A) 1 (B) 3 (C) 5 (D) 2

FN0092

18. Let $f: [1, \infty) \rightarrow [3, \infty)$ be defined as $f(x) = (\log_2 x)^2 + 2(\log_2 x) + 3$. If $f^{-1}(x)$ is inverse of $f(x)$, then

- (A) $f^{-1}(x) = 2^{-1+\sqrt{x-2}}$ (B) $f^{-1}(x) = 2^{-1-\sqrt{x-2}}$
 (C) domain of $f^{-1}(x)$ is $[2, \infty)$ (D) domain of $f^{-1}(x)$ is $[3, \infty)$

FN0207

19. If $f(x) = ax + b$ and $f(f(f(x))) = 27x + 13$ where a and b are real numbers, then-

- (A) $a + b = 3$ (B) $a + b = 4$ (C) $f'(x) = 3$ (D) $f'(x) = -3$

FN0208

20. If $f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) = f(x) \forall x \in \mathbb{R}$, then period of function $f(x)$ is greater than-

FN0209

- (A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 3

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 & 2

$$\text{Let } f(x) = \begin{cases} x & ; x < 0 \\ 1-x & ; x \geq 0 \end{cases} \quad \& \quad g(x) = \begin{cases} x^2 & ; x < -1 \\ 2x+3 & ; -1 \leq x \leq 1 \\ x & ; x > 1 \end{cases}$$

On the basis of above information, answer the following questions :

1. Range of $f(x)$ is -

(A) $(-\infty, 1]$ (B) $(-\infty, \infty)$ (C) $(-\infty, 0]$ (D) $(-\infty, 2]$

FN0063

2. Range of $g(f(x))$ is -

(A) $(-\infty, \infty)$ (B) $[1, 3) \cup (3, \infty)$ (C) $[1, \infty)$ (D) $[0, \infty)$

FN0063

Paragraph for Question 3 & 4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying $f(2-x) = f(2+x)$ and $f(20-x) = f(x)$, $\forall x \in \mathbb{R}$.

On the basis of above information, answer the following questions :

3. If $f(0) = 5$, then minimum possible number of values of x satisfying $f(x) = 5$, for $x \in [0, 170]$ is-

FN0169

4. Graph of $y = f(x)$ is symmetrical about $x = k$. Find k -

FN0169

Paragraph for Question 5 to 7

Let $f : (-\infty, -2] \rightarrow [3, \infty)$ defined by $f(x) = x^2 + 2x + 3$.

On the basis of above information, answer the following questions :

5. $f^{-1}(x)$ is equal to -

FN0210

(A) $\sqrt{x+2} - 1$ (B) $\sqrt{x-2} - 1$ (C) $-\sqrt{x+2} - 1$ (D) $-\sqrt{x-2} - 1$

6. Range of the function $g(x) = f(2x)$ is equal to -

FN0210

(A) $[1, \infty)$ (B) $[2, \infty)$ (C) $[3, \infty)$ (D) $[4, \infty)$

7. Number of solutions of $f(x) = f^{-1}(x)$ is equal to -

FN0210

(A) 0 (B) 1 (C) 2 (D) 3

Matrix Match Type

8. **Column-I** **Column-II**
Number of integers in

- (A) Domain of $f(x) = \ln\{x\}$ (P) 0
(B) Domain of $f(x) = \sqrt{\sec(\sin x)} + \sqrt{\left[x + \frac{1}{x}\right]} + \sqrt{10 - [x]^2}$ (Q) 2
(C) Range of $f(x) = x^2 - 2x + 2, x \in [0, 2]$ (R) 3
(D) Range of $f(x) = \sqrt{25 - [x]^2}$ (S) less than 3
(T) more than 3

(where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively)

FN0095

9. Match the function mentioned in column-I with the respective classification given in column-II.
(where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively)

Column-I **Column-II**

- (A) $f: \mathbb{R} \rightarrow \mathbb{R}^+ f(x) = (e^{[x]})(e^{\{x\}})$ (P) one-one
(B) $f: (-\infty, -2) \cup (0, \infty) \rightarrow \mathbb{R} f(x) = \ln(x^2 + 2x)$ (Q) many-one
(C) $f: [-2, 2] \rightarrow [-1, 1] f(x) = \sin x$ (R) onto
(D) $f: \mathbb{R} \rightarrow \mathbb{R} f(x) = x^3 - 3x^2 + 3x - 7$ (S) periodic
(T) aperiodic

FN0096

10. **List-I** **List-II**

- f(x)** **Range**
(I) $\frac{\cos^2 x + \cos x + 2}{\cos^2 x + \cos x + 1}$ (P) $\left[0, \frac{7}{3}\right]$
(II) $\left| \frac{(\sqrt{\cos x} - \sqrt{\sin x})(\sqrt{\cos x} + \sqrt{\sin x})}{3(\cos x + \sin x)} \right|$ (Q) $\left[\frac{4}{3}, \frac{7}{3}\right]$
(III) $\frac{7}{3(x^6 + 2x^4 + 3x^2 + 1)}$ (R) $\left[0, \frac{1}{3}\right]$
(IV) $\log_8(x^2 + 2x + 2)$ (S) $[0, \infty)$

FN0153

Which of the following is correct ?

- (A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S (B) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P
(C) I $\rightarrow$ P; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ R (D) I $\rightarrow$ P; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ Q

EXERCISE (O-4)

Numerical Grid Type

1. The sum of integral values of the elements in the domain of $f(x) = \sqrt{\log_{\frac{1}{2}} |3-x|}$ is FN0116
2. The number of even integral value(s) in the range of the function $f(x) = \frac{\tan^2 x + 8 \tan x + 15}{1 + \tan^2 x}$ is FN0118
3. Number of integers in the range of the function $f(x) = \frac{x^3 + 2x^2 + 3x + 2}{x^3 + 2x^2 + 2x + 1}$; $x \in \mathbb{R}^*$ is : FN0211
4. Let f be a one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statements is true and the remaining two are false.
 $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. Determine $f^{-1}(1)$ FN0138
5. Let ' f ' be an even periodic function with period '4' such that $f(x) = 2^x - 1$, $0 \leq x \leq 2$. The number of solutions of the equation $f(x) = 1$ in $[-10, 20]$ are FN0150
6. If $f(x) = a \log\left(\frac{1+x}{1-x}\right) + bx^3 + c \sin x + 5$ and $f(\log_3 2) = 4$, then $f\left(\log_3\left(\frac{1}{2}\right)\right)$ is equal to FN0155
7. Let $f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$. If $f(x)$ is divided by $x^3 - x$ then the remainder is some function of x say $g(x)$. Find the value of $g(10)$. FN0163
8. Let $f(x) = \frac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$ FN0165
9. The function $f(x)$ has the property that for each real number x in its domain, $1/x$ is also in its domain and $f(x) + f(1/x) = x$. The largest set of real numbers that can be in the domain of $f(x)$ contains k elements then k is equal to ? FN0170
10. A function ' f ' is defined for all real numbers and satisfies $f(2+x) = f(2-x)$ and $f(7+x) = f(7-x)$ for all real x . If $x = 0$ is a root of $f(x) = 0$, then find least number of roots of $f(x) = 0$ for $x \in [-1000, 1000]$. FN0212

EXERCISE (JM)

1. If $a \in \mathbb{R}$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

[JEE(Main)-2014]

- (1) $(-1, 0) \cup (0, 1)$ (2) $(1, 2)$
(3) $(-2, -1)$ (4) $(-\infty, -2) \cup (2, \infty)$

FN0173

2. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$, and

[JEE(Main)-2016]

$S = \{x \in \mathbb{R} : f(x) = f(-x)\}$; then S :

- (1) contains more than two elements. (2) is an empty set.
(3) contains exactly one element (4) contains exactly two elements

FN0174

3. The function $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$, is :

[JEE(Main)-2017]

- (1) neither injective nor surjective. (2) invertible.
(3) injective but not surjective. (4) surjective but not injective

FN0175

4. Let $S = \{x \in \mathbb{R} : x \geq 0 \text{ and } 2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$. Then S :

[JEE(Main)-2018]

- (1) contains exactly one element. (2) contains exactly two elements.
(3) contains exactly four elements. (4) is an empty set.

FN0176

5. For $x \in \mathbb{R} - \{0, 1\}$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function,

$J(x)$ satisfies $(f_2 \circ f_1)(x) = f_3(x)$ then $J(x)$ is equal to :-

[JEE(Main)-2019]

- (1) $f_3(x)$ (2) $f_1(x)$ (3) $f_2(x)$ (4) $\frac{1}{x} f_3(x)$

FN0177

6. Let $\mathbb{N}$ be the set of natural numbers and two functions f and g be defined as $f, g : \mathbb{N} \rightarrow \mathbb{N}$

such that : $f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$ and $g(n) = n - (-1)^n$. The fog is :

[JEE(Main)-2019]

- (1) Both one-one and onto (2) One-one but not onto
(3) Neither one-one nor onto (4) onto but not one-one

FN0178

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x}{1+x^2}$, $x \in \mathbb{R}$. Then the range of f is : [JEE(Main)-2019]

- (1) $(-1, 1) - \{0\}$ (2) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (3) $\mathbb{R} - \left[-\frac{1}{2}, \frac{1}{2}\right]$ (4) $\mathbb{R} - [-1, 1]$

FN0179

8. The number of functions f from $\{1, 2, 3, \dots, 20\}$ onto $\{1, 2, 3, \dots, 20\}$ such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4, is :- [JEE(Main)-2019]

- (1) $(15)! \times 6!$ (2) $5^6 \times 15$ (3) $5! \times 6!$ (4) $6^5 \times (15)!$

FN0180

9. If $f(x) = \log_e \left(\frac{1-x}{1+x} \right)$, $|x| < 1$, then $f\left(\frac{2x}{1+x^2}\right)$ is equal to : [JEE(Main)-2019]

- (1) $2f(x)$ (2) $2f(x^2)$ (3) $(f(x))^2$ (4) $-2f(x)$

FN0181

10. Let $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$, where the function f satisfies $f(x+y) = f(x)f(y)$ for all natural numbers

x, y and $f(1) = 2$. then the natural number 'a' is [JEE(Main)-2019]

- (1) 4 (2) 3 (3) 16 (4) 2

FN0182

11. Let $f(x) = x^2$, $x \in \mathbb{R}$. For any $A \subseteq \mathbb{R}$, define $g(A) = \{x \in \mathbb{R}, f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true ? [JEE(Main)-2019]

- (1) $f(g(S)) \neq f(S)$ (2) $f(g(S)) = S$ (3) $g(f(S)) = g(S)$ (4) $g(f(S)) \neq S$

FN0183

12. The number of real roots of the equation $5 + |2^x - 1| = 2^x (2^x - 2)$ is : [JEE(Main)-2019]

- (1) 2 (2) 3 (3) 4 (4) 1

FN0184

13. For $x \in \left(0, \frac{3}{2}\right)$, let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$. If $\phi(x) = ((h \circ f) \circ g)(x)$, then $\phi = \left(\frac{\pi}{3}\right)$ is equal to : [JEE(Main)-2019]

- (1) $\tan \frac{\pi}{12}$ (2) $\tan \frac{7\pi}{12}$ (3) $\tan \frac{11\pi}{12}$ (4) $\tan \frac{5\pi}{12}$

FN0185

14. For $x \in \mathbb{R}$, let $[x]$ denote the greatest integer $\leq x$, then the sum of the series

$\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \left[-\frac{1}{3} - \frac{2}{100}\right] + \dots + \left[-\frac{1}{3} - \frac{99}{100}\right]$ is [JEE(Main)-2019]

- (1) -153 (2) -133 (3) -131 (4) -135

FN0186

15. If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to [JEE(Main)-2020]

- (1) $\frac{3}{2}$ (2) $-\frac{1}{2}$ (3) $-\frac{3}{2}$ (4) $\frac{1}{2}$

FN0187

16. Let $f : (1,3) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$.

Then the range of f is

[JEE(Main)-2020]

- (1) $\left(\frac{3}{5}, \frac{4}{5}\right)$ (2) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$ (3) $\left(\frac{2}{5}, \frac{4}{5}\right]$ (4) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

FN0188

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that for all $x \in \mathbb{R}$ ($2^{1+x} + 2^{1-x}$), $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is [JEE(Main)-2020]

- (1) 0 (2) 3 (3) 2 (4) 4

FN0189

18. The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1,1)$, is [JEE(Main)-2020]

- (1) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1-x}{1+x}\right)$ (2) $\frac{1}{4} \log_e \left(\frac{1-x}{1+x}\right)$
(3) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1+x}{1-x}\right)$ (4) $\frac{1}{4} \log_e \left(\frac{1+x}{1-x}\right)$

FN0190

EXERCISE (JA)

1. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}, \quad g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}, \text{ then } (f - g)(x) \text{ is -}$$

- (A) one-one and onto (B) neither one-one nor onto
(C) one-one but not onto (D) onto but not one-one [JEE 2005 (Scr.)]

FN0191

2. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is- [JEE 2011, 3, (-1)]

- (A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ (B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
(C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ (D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

FN0192

3. The function $f : [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is : [JEE 2012, 3, (-1)]

- (A) one-one and onto (B) onto but not one-one
(C) one-one but not onto (D) neither one-one nor onto

FN0193

4. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then the value(s) of

$$f\left(\frac{1}{3}\right) \text{ is (are) - [JEE 2012, 4]}$$

- (A) $1 - \sqrt{\frac{3}{2}}$ (B) $1 + \sqrt{\frac{3}{2}}$ (C) $1 - \sqrt{\frac{2}{3}}$ (D) $1 + \sqrt{\frac{2}{3}}$

FN0194

5. Let the function $f : [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$ [JEE(Advanced)-2020]

$$\text{Then the value of } f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \dots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) \text{ is } \underline{\hspace{2cm}}$$

FN0198

6. Let $|M|$ denote the determinant of a square matrix M . Let $g : \left[0, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be the function defined

$$\text{by } g(\theta) = \sqrt{f(\theta)-1} + \sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$$

$$\text{where } f(\theta) = \frac{1}{2} \begin{vmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{vmatrix} + \begin{vmatrix} \sin \pi & \cos\left(\theta + \frac{\pi}{4}\right) & \tan\left(\theta - \frac{\pi}{4}\right) \\ \sin\left(\theta - \frac{\pi}{4}\right) & -\cos \frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right) \\ \cot\left(\theta + \frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan \pi \end{vmatrix}.$$

Let $p(x)$ be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2) = 2 - \sqrt{2}$. Then, which of the following is / are TRUE? **FN0213**

- (A) $p\left(\frac{3+\sqrt{2}}{4}\right) < 0$ (B) $p\left(\frac{1+3\sqrt{2}}{4}\right) > 0$ **[JEE(Advanced)-2022]**
 (C) $p\left(\frac{5\sqrt{2}-1}{4}\right) > 0$ (D) $p\left(\frac{5-\sqrt{2}}{4}\right) < 0$

ANSWER KEY

Do yourself - 1

1. (C,D) 2. (A,D) 3. $R = \{(13, 12), (15, 12), (15, 14), (17, 12), (17, 14), (17, 16)\}$
 4. (B) 5. (A) 6. (A,D) 7. $A = \{1, 2\}; B = \{p, q, r\}$

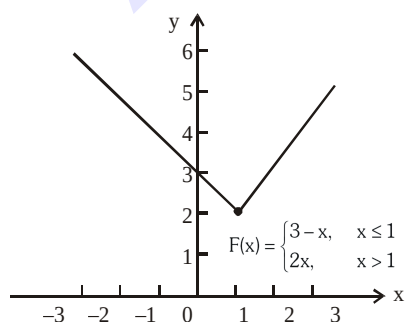
Do yourself - 2

1. $x \in (-\infty, -3) \cup (-1, 1] \cup [2, \infty)$ 2. $x \in (-\infty, -4) \cup [-4, 0] \cup [1, 2) \cup [5, \infty)$
 3. $x \in [-2, 0] \cup [2, \infty)$
 4. (i) $-\infty < x \leq 0$ (ii) $-1 \leq x \leq 1$ (iii) $x \in \mathbb{R}$ (iv) $x \in \mathbb{R} - \{-1, 0, 1\}$
 (v) $-\infty < x \leq 1$ or $3 \leq x < \infty$ (vi) $-\infty < x < 1$ or $2 < x < \infty$
 5. $\left(-\infty, -\frac{1}{2}\right] \cup (0, 1) \cup (2, \infty)$
 6. (i) $x \in (0, \infty)$ (ii) $x \in (-\infty, 0) \cup (4, \infty)$
 7. (i) $0 < x < 1$ or $1 < x < \infty$ (ii) $-2 \leq x < 0$ or $0 < x < 1$ (iii) $\frac{3}{2} < x < 2$ or $2 < x < \infty$
 8. (i) $x \in \phi$ (ii) $x \in \mathbb{R}$ (iii) $x \in [0, \infty)$ (iv) $x \in (-\infty, 0)$
 9. (i) $4 \leq x \leq 6$ (ii) $2 < x < 3$
 10. (i) $-1 < x < 0$ or $1 < x < 2$; $2 < x < \infty$ (ii) $2k\pi < x < (2k+1)\pi$, where k is an integer
 11. (B) 12. (A)

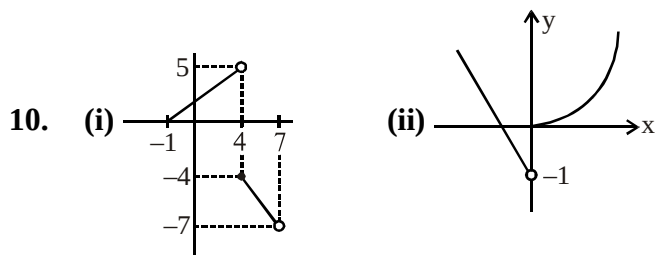
Do yourself - 3

1. (i) $(-\infty, \infty)$ (ii) $[2, 20]$
 2. (i) $[-9, \infty)$ (ii) $[-8, 16]$ (iii) $[-9, \infty)$ (iv) $[-8, \infty)$
 3. (i) $[1, 9]$ (ii) $[1, 5]$ (iii) $[6, 12]$
 4. (i) $\left(0, \frac{4}{3}\right]$ (ii) $\left(-\infty, \frac{-16}{9}\right] \cup (0, \infty)$ 5. 4
 6. (i) $[2, \infty)$ (ii) $\left[\frac{1}{11}, \frac{1}{5}\right]$ 7. $\left[\frac{1}{3}, 1\right]$

8.



9. (i) -2 (ii) -3, 1 (iii) $[-3, 3], [-2, 3]$ (iv) $[-1, 3]$ (v) (B) (vi) (A,D)



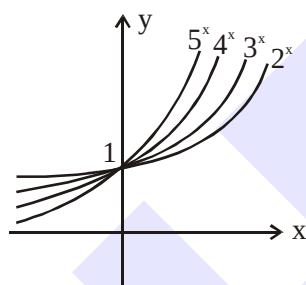
Range : $(-7, -4] \cup [0, 5)$ Range : $(-1, \infty)$

11. (i) -2, 4 (ii) -2, 2 (iii) -3, 4 (iv) $[0, 4]$ (v) $[-4, 4], [-2, 3]$ (vi) $[-4, 3], [0.5, 4]$
12. (i) $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$ (ii) $[-2, -1) \cup [1, 3]$ (iii) $[-3, -2] \cup [-1, 1]$ (iv) $\left(\frac{5}{3}, 3\right) - \{2\}$
13. (C) 14. (B)

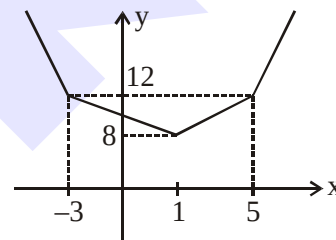
Do yourself - 4

1. (B)

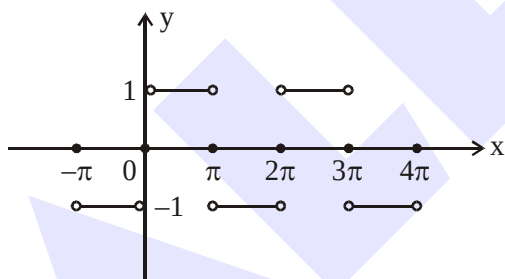
2.



3.



4.

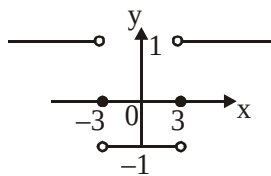


5. (i) $y = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$ (ii) $y = \begin{cases} 2, & 0 \leq x < 1 \text{ and } 2 \leq x < 3 \\ 0, & 1 \leq x < 2 \text{ and } 3 \leq x \leq 4 \end{cases}$

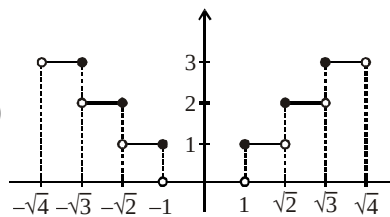
Do yourself - 5

1. (i) $x \in [-10, -9) \cup [10, 11)$ (ii) $x \in [-1, 0) \cup [2, 3)$
 (iii) $x \in [5, \infty)$ (iv) $x \in (-\infty, -3)$
2. $x \in (-\infty, 7) \cup [11, \infty)$
3. (i) $x \in [3, 4)$ (ii) 7 (iii) 17
4. 3897 5. $x \in (-\infty, -19) \cup (19, \infty)$ 6. $R \in \left[\frac{3}{5}, \frac{2}{3}\right)$

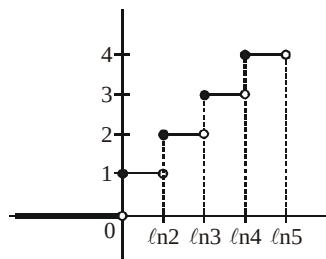
7. (i)



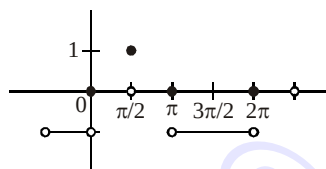
(ii)



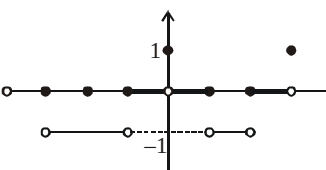
(iii)



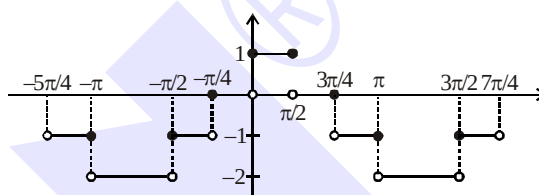
(iv)



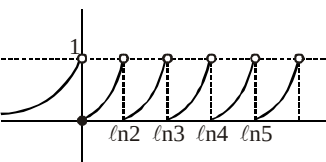
(v)



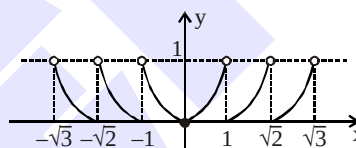
(vi)



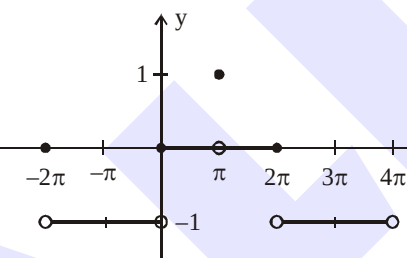
8. (i)



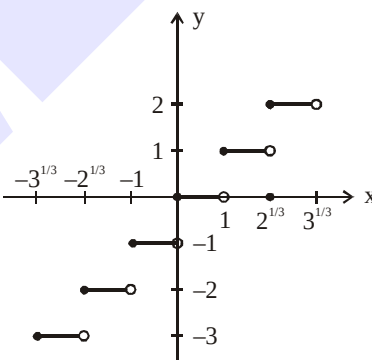
(ii)



9. (i)



(ii)



10. (i) $x \in \left\{ \frac{7}{6}, \frac{5}{2}, \frac{23}{6} \right\}$ (ii) $x = 5/3$

11. (i) $x \in \{0\}$ (ii) $x \in [0, 1)$

(iii) $x \in \left\{ -5, -\frac{41}{7}, -\frac{47}{7}, -\frac{53}{7}, -\frac{59}{7}, -\frac{65}{7}, -\frac{71}{7} \right\}$

12. $x = 0$ or $5/3$

13. D

14. C

15. (A) $\rightarrow (p, q, r, s)$, (B) $\rightarrow (p, q)$, (C) $\rightarrow (t)$, (D) $\rightarrow (r)$

Do yourself - 6

1. (i) $x \in \mathbb{R}$ (ii) $x \in (-\infty, -3) \cup [2, \infty)$ 2. $(-1, 0) \cup (0, 1]$
3. (i) $\left[-\frac{5\pi}{4}, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$ (ii) $\left(-4, -\frac{1}{2}\right) \cup (2, \infty)$
- (iii) $(-\infty, -3]$ (iv) $(-\infty, -1) \cup [0, \infty)$
4. (i) $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \leq 4)$ (ii) $\left(0, \frac{1}{100}\right) \cup \left(\frac{1}{100}, \frac{1}{\sqrt{10}}\right)$
5. (i) $(-3, -1] \cup \{0\} \cup [1, 3)$ (ii) $\{4\} \cup [5, \infty)$ (iii) $(1, 2) \cup (2, 5/2)$
6. (i) $(0, 1/4) \cup (3/4, 1) \cup \{x : x \in \mathbb{N}, x \geq 2\}$ (ii) $\mathbb{R} - \left\{-\frac{1}{2}, 0\right\}$
7. (i) $x \in (0, \infty) - \{3\}$ (ii) $x \in (-2, 3)$
8. (i) $x \in (2, \infty)$ (ii) $x \in (0, 1) \cup (1, \infty)$ (iii) $x \in (2, 3) \cup (3, \infty)$
9. D 10. C

Do yourself - 7

1. (i) $2K\pi \leq x \leq 2K\pi + \pi$ where $K \in \mathbb{I}$ (ii) $[-3/2, -1]$
2. (i) Range : $[-1/3, 3]$, Domain = $[4, 7]$; (ii) Range $[-1, 9]$ and domain $[11, 14]$
3. (i) $(-\infty, \infty)$ (ii) $\left[-1, \log_2 \frac{3}{4}\right]$ 4. (i) $\mathbb{R} - \{0\}$ (ii) $(0, 4/3]$ (iii) $[1/3, 3]$
5. (i) $[1, \infty)$ (ii) $\mathbb{R}$ (iii) $[0, \infty)$ 6. (i) $\mathbb{R} - \left\{\frac{5}{4}\right\}$ (ii) $\mathbb{R} - \left\{\frac{7}{3}\right\}$
7. (i) $\left[\frac{55}{8}, 13\right]$ (ii) $(-8, 8)$ (iii) $[-1, 1]$
8. (i) $\mathbb{R} - \left\{\frac{1}{4}, \frac{-1}{11}\right\}$ (ii) $(-\infty, -1] \cup (0, \infty) - \left\{\frac{-3}{2}\right\}$ (iii) $\left(-\infty, \frac{-4}{25}\right] \cup (0, \infty)$
9. (i) $\left[\frac{1}{2}, \infty\right)$ (ii) $[-1, 1]$ (iii) $[-2, 2]$ (iv) $[-1, 1]$
10. (i) $\left[\frac{1}{2}, 1\right]$ (ii) $\left[\ln \frac{11}{3}, \infty\right)$ 11. (i) $(-1, 1)$ (ii) $\mathbb{R}$
12. (i) $D : x \in \mathbb{R} \quad \mathbb{R} : [0, 2]$ (ii) $D = \mathbb{R}$; range $[-1, 1]$
- (iii) $D : \{x \mid x \in \mathbb{R}; x \neq -3; x \neq 2\} \quad \mathbb{R} : \{f(x) \mid f(x) \in \mathbb{R}, f(x) \neq 1/5; f(x) \neq 1\}$
13. (i) $D : \mathbb{R}, \mathbb{R} : (-1, 1)$ (ii) $D : -1 \leq x \leq 2 \quad \mathbb{R} : [\sqrt{3}, \sqrt{6}]$
- (iii) $D : [-4, \infty) - \{5\}; \mathbb{R} : \left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right)$
14. (i) $\mathbb{R} - \{1\}$ (ii) $[3, 7]$ (iii) $[-1, 1]$ (iv) $[1/3, 1)$

15. $(-\infty, \infty)$

16. B

17. C

18. A, C

19. D

20. D

21. C

22. 5049

Do yourself - 8

1. A

2. C

3. A, B, C

4. (i) no (ii) no (iii) yes

5. $\left[0, \frac{1}{2}\right]$ 6. (i) $y = \frac{1}{x}$ (ii) $y = \frac{1}{x^2}$ 7. (i) $y = \log(10 - 10^x)$, $-\infty < x < 1$ (ii) $y = x/3$ when $-\infty < x < 0$ & $y = x$ when $0 \leq x < +\infty$

8. (i) are identical (ii) are not identical 9. (i) are identical (ii) are not identical

Do yourself - 9

1. A

2. C

3. A

4. (i) $-3/4$; (ii) 64

5. 401

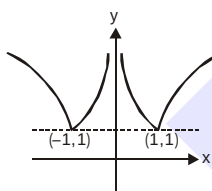
6. (i) 30

(ii) 5050

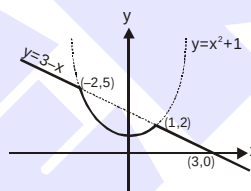
7. $f(x) = 2x^2$ **Do yourself - 10**

1. D

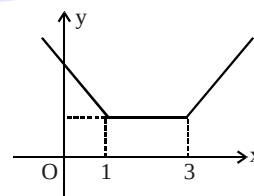
2. (i)



(ii)

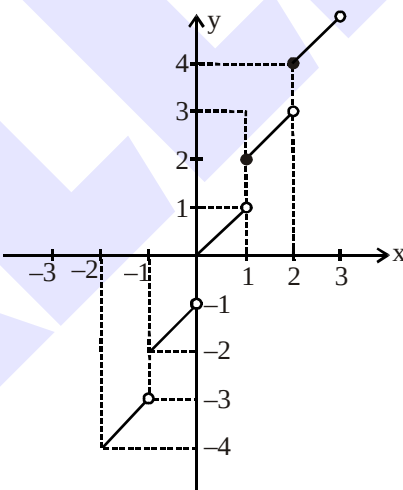


(iii)

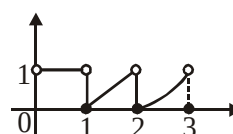


3. D

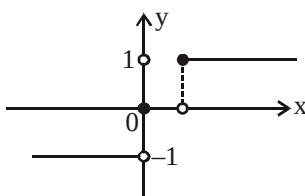
4. (i)



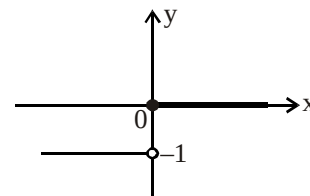
(ii)



(iii)



(iv)



5. (A)-S; (B)-R; (C)-P; (D)-Q

Do yourself - 11

1. A 2. A 3. B 4. A 5. (i) not onto (ii) yes
6. D 7. B
8. (i) neither surjective nor injective (ii) surjective but not injective
(iii) neither injective nor surjective
9. $(A) \rightarrow (P, R, T); (B) \rightarrow (Q, R); (C) \rightarrow (Q, S); (D) \rightarrow (P, R, T)$

Do yourself - 12

1. D 2. B 3. C
4. (i) (a) 0 (b) 0 (c) 0 (d) 0 (e) 0 (f) 0

$$(ii) \begin{cases} x+2, & 0 \leq x < 1 \\ x+1, & 1 \leq x < 2 \\ x, & 2 \leq x < 3 \end{cases}$$

5. (i) domain is $x \geq 0$; range $[-1, 1]$; (ii) domain $2k\pi \leq x \leq 2k\pi + \pi$; range $[0, 1]$
(iii) domain $x \in \mathbb{R}$; range $[-\sin 1, \sin 1]$ (iv) domain is $0 \leq x \leq 1$; range is $[0, 1]$

$$6. \frac{(x+3)^{10}}{(x+3)^{10}+1}, \text{ domain is } \mathbb{R}, \frac{1024}{1025}$$

$$7. g(x) = \begin{cases} \frac{1}{x^2} & \text{if } 0 < x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

$$8. fog(x) = \begin{cases} -(1+x), & -1 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}; gof(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 3-x, & 1 \leq x \leq 2 \\ x-1, & 2 < x \leq 3 \\ 5-x, & 3 < x \leq 4 \end{cases}$$

$$10. (gof)(x) = \begin{cases} x & \text{if } x \leq 0 \\ -x^2 & \text{if } 0 < x < 1 \\ 1-x^2 & \text{if } x \geq 1 \end{cases}; (fog)(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1+x & \text{if } 0 \leq x < 1 \\ x & \text{if } x \geq 1 \end{cases}$$

$$11. 2$$

Do yourself - 13

1. B 2. 3 3. B 4. B 5. B

$$6. A 7. D 8. (i) f^{-1}(x) = \begin{cases} -\sqrt{-x}, & -1 \leq x \leq 0 \\ \sqrt{x}, & 0 \leq x \leq 1 \end{cases} \quad (ii) y = -2 + e^{x-1}$$

$$9. (i) \frac{e^x - e^{-x}}{2}; \quad (ii) \frac{\log_2 x}{\log_2 x - 1}; \quad (iii) \frac{1}{2} \log \frac{1+x}{1-x}$$

$$10. x = 10; f^{-1}(x) = 10^{\log_2(x-8)}$$

Do yourself - 14

- (i) even (ii) odd (iii) odd (iv) neither even nor odd
- (i) odd (ii) even (iii) neither odd nor even (iv) odd
(v) neither odd nor even (vi) even (vii) even (viii) even
- 3

Do yourself - 15

- A, B, C
- C
- 9
- 0.86
- (i) 2π (ii) 1 (iii) π
- $(A) \rightarrow (P, R, T); (B) \rightarrow (Q, R, T); (C) \rightarrow (Q, R, T); (D) \rightarrow (P, R, T)$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	B	B	A	C	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	D	B	A	B	B	B	B	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	A	B	C	B	A	D	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B, C	A, B, C, D	A, B, C	A, C	A, B, C	B, C, D	A, C	C, D	A, B, C, D	B, D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B, C, D	A, B, C	A, D	A, B, C	A, B, C, D	A, C	B, C	A, D	B, C	A, B, C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7			
Ans.	A	C	22.00	18.00	D	C	A			
Q.8	A	B	C	D	Q.9	A	B	C	D	10
	P, S	R	Q, S	T		P, R, T	Q, R, T	Q, R, T	P, R, T	A

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	6	9	0	$f^{-1}(1) = y$	15	6	21	1002.50	2	401

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	4	2	1	4	2	1	1	2
Que.	11	12	13	14	15	16	17	18		
Ans.	3	4	3	2	2	4	2	3		

EXERCISE # JEE-ADVANCED

1. A
2. A
3. B
4. (zero marks to all) Reason : (as we are getting two f images for one x in the domain)
5. 19.00
6. A,C

EXERCISE (O-1)

Straight Objective Type

1. The domain of the function $\sin^{-1}\left(\log_2\left(\frac{x}{3}\right)\right)$ is-
(A) $\left[\frac{1}{2}, 3\right]$ (B) $\left[\frac{1}{2}, 3\right]$ (C) $\left[\frac{3}{2}, 6\right]$ (D) $\left[\frac{1}{2}, 2\right]$
IT0001
2. Domain of the function $f(x) = \log_e \cos^{-1}\{\sqrt{x}\}$ is, where $\{.\}$ represents fractional part function -
(A) $x \in \mathbb{R}$ (B) $x \in [0, \infty)$ (C) $x \in (0, \infty)$ (D) $x \in \mathbb{R} - \{x \mid x \in \mathbb{I}\}$
IT0002
3. The range of the function $f(x) = \sin^{-1}(\log_2(-x^2 + 2x + 3))$ is -
(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right]$ (C) $\left[0, \frac{\pi}{2}\right]$ (D) $[-1, 1]$
IT0016
4. Range of $f(x) = \cot^{-1}(\log_e(1 - x^2))$ is -
(A) $(0, \pi)$ (B) $\left(0, \frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{2}, \pi\right)$ (D) $\left(0, \frac{\pi}{2}\right]$
IT0017
5. Let $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then the value of $f(1) + f(2)$ is $k\pi$ where k is equal to
(A) 0 (B) 1 (C) 2 (D) 3
IT0106
6. The function $f(x) = \cot^{-1}\sqrt{(x+3)x} + \cos^{-1}\sqrt{x^2+3x+1}$ is defined on the set S , where S is equal to:
(A) $\{0, 3\}$ (B) $(0, 3)$ (C) $\{0, -3\}$ (D) $[-3, 0]$
IT0107
7. Range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$ is
(A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (D) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$
IT0108
8. $\operatorname{cosec}^{-1}(\cos x)$ exists if
(A) $x \in [-1, 1]$ (B) $x \in \mathbb{R}$
(C) x is an odd multiple of $\frac{\pi}{2}$ (D) x is a multiple of π
IT0109

9. The value of $\cot^{-1} \left\{ \frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}} \right\}$, where $\frac{\pi}{2} < x < \pi$, is: **IT0110**

- (A) $\pi - \frac{x}{2}$ (B) $\frac{\pi}{2} + \frac{x}{2}$ (C) $\frac{x}{2}$ (D) $2\pi - \frac{x}{2}$

10. The inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ holds for **IT0111**

- (A) $x \in (2 - \sqrt{9-2\pi}, 2 + \sqrt{9-2\pi})$ (B) $x > 2 + \sqrt{9-2\pi}$

- (C) $x < 2 - \sqrt{9-2\pi}$ (D) $x \in \phi$

11. Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7)$. If y simplifies to $a\pi + b$, then find $(a - b)$. **IT0041**

- (A) 50 (B) 48 (C) 53 (D) 42

12. Show that the roots r , s and t of the cubic $x(x-2)(3x-7) = 2$, are real and positive. Also compute the value of $\tan^{-1}(r) + \tan^{-1}(s) + \tan^{-1}(t)$. **IT0073**

- (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) 2π

13. The complete solution set of the inequality $[\cot^{-1} x]^2 - 6[\cot^{-1} x] + 9 \leq 0$, where $[.]$ denotes greatest integer function, is **IT0130**

- (A) $(-\infty, \cot 3]$ (B) $[\cot 3, \cot 2]$ (C) $[\cot 3, \infty)$ (D) $(-\infty, \cot 2]$

14. If $2 \leq a < 3$, then the value of $\cos^{-1} \cos[a] + \operatorname{cosec}^{-1} \operatorname{cosec}[a] + \cot^{-1} \cot[a]$, (where $[.]$ denotes greatest integer less than equal to x) is equal to

- (A) $2 - \pi$ (B) $2 + \pi$ (C) π (D) 6

IT0005

15. The value of $\tan^2(\sec^{-1} 3) + \cot^2(\operatorname{cosec}^{-1} 4)$ is -

- (A) 9 (B) 16 (C) 25 (D) 23

IT0003

16. $\tan^{-1} \left(1 - x^2 - \frac{1}{x^2} \right) + \sin^{-1} \left(x^2 + \frac{1}{x^2} - 1 \right)$ (where $x \neq 0$) is equal to

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) π

IT0004

17. If $x > 0$ $\cos^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{16}{x}\right)$ then x is -

- (A) 12 (B) 16 (C) 20 (D) 320

IT0006

18. Number of integral ordered pairs (a, b) for which $\sin^{-1}(1 + b + b^2 + \dots \infty) + \cos^{-1}\left(a - \frac{a^2}{3} + \frac{a^3}{9} - \dots \infty\right) = \frac{\pi}{2}$ is-

- (A) 0 (B) 4 (C) 9 (D) Infinitely many

IT0007

19. If $\cos^{-1}(2x^2 - 1) = 2\pi - 2\cos^{-1}x$, then -

- (A) $x \in [-1, 0]$ (B) $x \in [0, 1]$ (C) $x \in \left[0, \frac{1}{\sqrt{2}}\right]$ (D) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

IT0008

20. The value of $\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$ is-

- (A) 5 (B) 6 (C) 0 (D) 10

IT0018

21. Number of solution(s) of the equation $\cos^{-1}\sqrt{x} - \sin^{-1}\sqrt{x-1} + \cos^{-1}\sqrt{1-x} - \sin^{-1}\frac{1}{\sqrt{x}} = \frac{\pi}{2}$ is -

- (A) 0 (B) 1 (C) 2 (D) 4

IT0019

22. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy \cos\alpha + y^2$ is equal to-

- (A) $2 \sin 2\alpha$ (B) 4 (C) $4 \sin^2 \alpha$ (D) $-4 \sin^2 \alpha$

IT0112

23. $\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x\right)$, $x \neq 0$ is equal to

- (A) x (B) $2x$ (C) $\frac{2}{x}$ (D) $\frac{x}{2}$

IT0113

24. If $\alpha = 2 \arctan\left(\frac{1+x}{1-x}\right)$ & $\beta = \arcsin\left(\frac{1-x^2}{1+x^2}\right)$ for $0 < x < 1$, then prove that $\alpha + \beta = \pi$, what the value of $\alpha + \beta$ will be if $x > 1$.

IT0062

- (A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) 2π

25. The numerical value of $\tan\left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4}\right)$ is IT0126

- (A) $\frac{-7}{17}$ (B) $\frac{7}{17}$ (C) $\frac{17}{7}$ (D) $-\frac{2}{3}$

26. If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then x equals

- (A) 1/2 (B) 1 (C) -1/2 (D) -1 IT0127

27. The number of solution(s) of the equation, $\sin^{-1}x + \cos^{-1}(1-x) = \sin^{-1}(-x)$, is IT0128

- (A) 0 (B) 1 (C) 2 (D) 3

28. $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{r((r+1)!)}{(r+1) + ((r+1)!)^2}\right)$ is equal to -

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\cot^{-1}3$ (D) $\tan^{-1}2$

IT0020

29. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{2r+1}{r^4 + 2r^3 + r^2 + 1}$ is equal to -

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{8}$

IT0009

30. If $0 < x < 1$, then $\sqrt{1+x^2} [\{x \cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2} =$

- (A) $\frac{x}{\sqrt{1+x^2}}$ (B) x (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

IT0091

EXERCISE (O-2)

Multiple Correct Answer Type

- Domain of $f(x) = \cos^{-1} x + \cot^{-1} x + \operatorname{cosec}^{-1} x$ is **IT0100**
 (A) $[-1, 1]$ (B) $\mathbb{R}$
 (C) $(-\infty, -1] \cup [1, \infty)$ (D) $\{-1, 1\}$
- Domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ for real valued 'x' is: **IT0101**
 (A) $\left[-\frac{1}{4}, \frac{1}{2}\right]$ (B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (C) $\left(-\frac{1}{2}, \frac{1}{9}\right)$ (D) $\left[-\frac{1}{4}, \frac{1}{4}\right]$
- The domain of definition of $f(x) = \sin^{-1}(|x-1|-2)$ is: **IT0102**
 (A) $[-2, 0] \cup [2, 4]$ (B) $(-2, 0) \cup (2, 4)$
 (C) $[-2, 0] \cup [1, 3]$ (D) $(-2, 0) \cup (1, 3)$
- If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$, then $\cos^{-1} x + \cos^{-1} y$ is equal to **IT0103**
 (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) π
- The values of 'x' satisfying the equation $\sin^{-1}(x^2 - 5x + 7) = 2\tan^{-1}1$ is (are) **IT0104**
 (A) 2 (B) 3 (C) 4 (D) 5
- Consider the function $f(x) = e^x$ and $g(x) = \sin^{-1}x$, then which of the following is/are necessarily true. **IT0105**
 (A) Domain of $g \circ f$ = Domain of f (B) Range of $g \circ f \subset$ Range of g
 (C) Domain of $g \circ f$ is $(-\infty, 0]$ (D) Range of $g \circ f$ is $\left[-\frac{\pi}{2}, 0\right)$
- Let $f(x) = \sin^{-1}(\tan x) + \cos^{-1}(\cot x)$ then **IT0013**
 (A) $f(x) = \frac{\pi}{2}$ wherever defined (B) domain of $f(x)$ is $x = n\pi \pm \frac{\pi}{4}, n \in \mathbb{I}$
 (C) period of $f(x)$ is $\frac{\pi}{2}$ (D) $f(x)$ is many one function

IT0011

8. Which of the following is/are correct ?

(A) $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$

(B) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \operatorname{sgn}(e^x)$ then $f(x)$ is an into function.

(C) If $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(x) = \sin x + x$ then $f(x)$ is an odd function.

(D) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \frac{e^x}{e^{[x]}}$ then $f(x)$ is a periodic function .

(where $[.]$ represents greatest integer function)

IT0012

9. If $\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + \cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) = \frac{p}{q}$, where p & q are relatively prime then digit at units

place of $(p - q)^{2k+1}$, $k \in \mathbb{N}$, can be -

(A) 1

(B) 3

(C) 7

(D) 9

IT0010

10. If $\frac{1}{2}\sin^{-1}\left(\frac{3\sin 2\theta}{5+4\cos 2\theta}\right) = \frac{\pi}{4}$, then $\tan\theta$ can not be equal to

IT0105

(A) 1

(B) 2

(C) 3

(D) 4

11. Let $f(x) = \begin{cases} x^2 - 4 & \text{if } |x| \leq 3 \\ 5\operatorname{sgn}|x-3| & \text{if } |x| > 3 \end{cases}$ and $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ for all $x \in \mathbb{R}$, then which of the

following is wrong ?

(where $\operatorname{sgn}(x)$ denotes signum function of x)

(A) $f(x)$ is an even function

(B) $g \circ f(x)$ is an even function

(C) $g(x)$ is an odd function

(D) $f \circ g(x)$ is an odd function

IT0024

12. Let $f(x) = \sqrt{\cos^{-1}\sqrt{1-x^2} - \sin^{-1}x}$ then which of the following statement/s is/are correct -

(A) Domain of $f(x)$ is $[-1, 1]$

(B) Domain of $f(x)$ is $[0, 1]$

(C) Range of $f(x)$ is $\{0\}$

(D) Range of $f(x)$ is $[0, \sqrt{\pi}]$

IT0021

13. If $x < 0$, then value of $\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$ is equal to

IT0114

(A) $\frac{\pi}{2}$

(B) $-\frac{\pi}{2}$

(C) 0

(D) $-\pi$

14. Number of solutions of equation $\tan^{-1}(e^{-x}) + \cot^{-1}(|\ln x|) = \pi/2$ is :

IT0114

(A) 0

(B) 1

(C) 3

(D) 2

15. Value of $3\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ is greater than

- (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{6}$

IT0015

16. If α is only real root of the equation $x^3 + (\cos 1)x^2 + (\sin 1)x + 1 = 0$, then $\left(\tan^{-1}\alpha + \tan^{-1}\frac{1}{\alpha}\right)$ cannot be equal to-

- (A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π

IT0023

17. If $\alpha = 2\tan^{-1}(\sqrt{3-2\sqrt{2}}) + \sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta = \cot^{-1}(\sqrt{3}-2) + \frac{1}{8}\sec^{-1}(-2)$ & $\gamma = \tan^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}\frac{1}{\sqrt{3}}$,

then

- (A) $\alpha = \beta$ (B) $\alpha + \beta = 3\gamma$ (C) $4(\beta - \gamma) = \alpha$ (D) $\beta = \gamma$

IT0022

18. If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right) + \sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) + \tan^{-1}y = \frac{2\pi}{3}$, then :

IT0115

- (A) maximum value of $x^2 + y^2$ is $\frac{49}{3}$ (B) maximum value of $x^2 + y^2$ is 4
(C) minimum value of $x^2 + y^2$ is $\frac{1}{2}$ (D) minimum value of $x^2 + y^2$ is 3

19. If $f(x) = \tan^{-1}\left(\frac{\sqrt{3x-3x^2}}{3\sqrt{3}+x^2}\right) + \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$, $0 \leq x \leq 3$, then range of $f(x)$ is

IT0115

- (A) $\left[0, \frac{\pi}{2}\right)$ (B) $\left[0, \frac{\pi}{4}\right]$ (C) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ (D) $\left[0, \frac{\pi}{3}\right]$

20. Integral value(s) of x for which inequality $\sin^{-1}\left(\sin\left(\frac{2e^x+3}{e^x+1}\right)\right) > \pi - \frac{5}{2}$ holds, is/are

IT0029

- (A) 1 (B) -2 (C) 3 (D) 5

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Consider a continuous function such that each image has atmost three preimage & atleast one image has exactly three preimages. This type of function is to be called as three-one function.

On the basis of above information, answer the following questions :

1. Which of the following function is a three-one function ?

(A) $|\ell n|x||$ (B) $e^{|x|}$ (C) $x^3 + 3x^2 - 7x + 6$ (D) $\cos(\cos^{-1}x)$

IT0025

2. If $f(x) = \sin^{-1}(\sin x)$ is a three-one function, then possible interval of x is -

(A) $[-\pi, \pi]$ (B) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (C) $(-2\pi, 0]$ (D) $\left[\frac{-3\pi}{2}, \frac{\pi}{2}\right]$

IT0025

3. If $f(x)$ is a three-one function such that $f(a) = f(b)$ (where $a \neq b$), then number of maximum possible values of b is -

(A) 1 (B) 2 (C) 3 (D) 4

IT0025

Paragraph for Question 4 to 5

If $\alpha = 2 \tan^{-1} \frac{1}{2} + \sin^{-1} \frac{3}{5}$, $\beta = \sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \cot^{-1} \frac{16}{63}$.

On the basis of above information, answer the following questions :

4. If $2\sin\alpha, \cos\beta$ are roots of $x^2 - ax + b = 0$, then -

IT0116

(A) $a = 1$ (B) $a = -1$ (C) $b = -2$ (D) $\frac{b}{a} = -2$

5. The range of $\cot^{-1}(x^2 + bx)$ is $(c, d]$, then

IT0117

(A) $c = 0$ (B) $d = \frac{3\pi}{4}$ (C) $c = \frac{\pi}{4}$ (D) $c + d = \frac{3\pi}{4}$

Paragraph for Question 6 to 7

Let a_r ($a_r > 0 \forall r \in \mathbb{N}$) be the r^{th} term of a GP such that $\sqrt{a_1 \cdot a_3 \cdot a_5 \cdot a_7} = (18a_1 + a_3)^2$ and $\sum_{r=1}^{\infty} \frac{1}{a_r} = \frac{3}{4}$.

On the basis of above information, answer the following questions :

6. The value of $\sum_{r=1}^{10} \tan\left(\sin^{-1}\left(\frac{a_{r+1}}{\sqrt{a_r^2 + a_{r+1}^2}}\right)\right)$ equals -

IT0118

7. The value of $\sin^{-1}\left(\sin\left(\frac{a_2}{a_1}\right)\right) + \tan^{-1}(\tan a_1)$ is equal to -

IT0119

Matrix Match Type

Match List-I with List-II and select the correct answer.

List-I

List-II

(P) The number of integral values of k for which the equation $\sin^{-1} x^2 + \tan^{-1} x^2 = 2k + 1$ has a solution

(1) 1

(Q) Let $\sum_{k=1}^{\infty} \cot^{-1} \left(\frac{k^2}{8} \right) = \frac{p}{q} \pi$ where $\frac{p}{q}$ is rational in its lowest form, then find $|q - p|$

(2) 2

(R) If $\tan^{-1} (\sin^2 \theta - 2 \sin \theta + 3) + \cot^{-1} (5^{\sec^2 y} + 1) = \frac{\pi}{2}$, then the value of $\cos^2 y - \sin \theta$

(3) 3

(S) Minimum positive integral value of x such that $f(x)$

(4) 0

is defined, $f(x) = \sqrt{\sec^{-1} \left(\frac{1 - |x|}{2} \right)}$

8. Which of the following is the only CORRECT combination?

IT0120

(A) (P), (3) (B) (S), (2) (C) (R), (2) (D) (Q), (1)

9. Which of the following is the only CORRECT combination?

IT0121

(A) (P), (1) (B) (Q), (1) (C) (R), (4) (D) (S), (2)

10. Match the column

IT0122

Column-I

Column-II

(A) Let a, b, c be three positive real numbers

(p) π

$$\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}},$$

then θ is equal to

(B) The value of the expression

$$\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) \text{ for } 0 < A < (\pi/4)$$

(q) $-\frac{\pi}{2}$

is equal to

(C) If $x < 0$, then $\frac{1}{2} \{ \cos^{-1} (2x^2 - 1) + 2 \cos^{-1} x \}$ is equal to

(r) $-\pi$

(D) The value of $\sin^{-1} \left(\frac{3}{5} \right) - \cos^{-1} \left(\frac{12}{13} \right) + \cos^{-1} \left(\frac{16}{65} \right)$ is equal to

(s) $\frac{\pi}{2}$

EXERCISE (O-4)

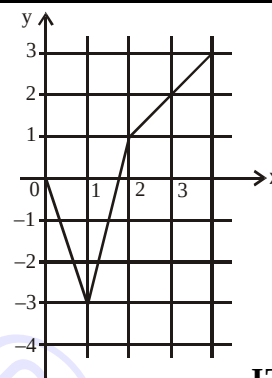
Numerical Grid Type

1. Given is a partial graph of an even periodic function

f whose period is 8. If $[*]$ denotes greatest integer

function then find the value of the expression.

$$f(-3) + 2|f(-1)| + \left[f\left(\frac{7}{8}\right) \right] + f(0) + \arccos(f(-2)) + f(-7) + f(20)$$



IT0030

2. If $\sin^{-1}\sin\left(\frac{10\pi}{3}\right) + \cos^{-1}\cos\frac{22\pi}{3} + \tan^{-1}\tan 10 = a\pi + b$, then $\left|\frac{3ab}{80}\right|$ is equal to

IT0026

3. Number of integral solutions of the equation $2\sin^{-1}\sqrt{x^2 - x + 1} + \cos^{-1}\sqrt{x^2 - x} = \frac{3\pi}{2}$ is

IT0027

4. The domain of the function $f(x) = \sin^{-1}\left(\frac{1+x^3}{2x^{3/2}}\right) + \sqrt{\sin(\sin x)} + \log_{(3\{x\}+1)}(x^2+1)$, contain how many integers where $\{.\}$ represents fractional part function, is:

IT0123

5. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, where $-1 \leq x, y, z \leq 1$, then find the value of $x^2 + y^2 + z^2 + 2xyz + \frac{3}{2}$

IT0124

6. If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$, $n \in \mathbb{N}$, then the maximum value of $\frac{n}{2}$ is:

IT0125

7. If $a > b > c > 0$, then find the value of : $\cot^{-1}\left(\frac{1+ab}{a-b}\right) + \cot^{-1}\left(\frac{1+bc}{b-c}\right) + \cot^{-1}\left(\frac{1+ca}{c-a}\right)$.

IT0045

8. If the sum $\sum_{n=1}^{10} \sum_{m=1}^{10} \tan^{-1}\left(\frac{m}{n}\right) = k\pi$, find the value of $\frac{k}{2}$.

IT0072

9. The number of real solutions of equation $\sqrt{1+\cos 2x} = \sqrt{2} \sin^{-1}(\sin x)$, $-10\pi \leq x \leq 10\pi$, is/are

IT0129

10. If α and β are the roots of the equation $x^2 + 5x - 49 = 0$, then find the value of $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$.

IT0044

EXERCISE (JM)

1. If $\cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}$ ($x > \frac{3}{4}$), then x is equal to : [JEE(Main)-2019]

(1) $\frac{\sqrt{145}}{12}$ (2) $\frac{\sqrt{145}}{10}$ (3) $\frac{\sqrt{146}}{12}$ (4) $\frac{\sqrt{145}}{11}$

IT0082

2. If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to: [JEE(Main)-2019]

(1) π (2) 7π (3) 0 (4) 10

IT0083

3. The value of $\cot\left(\sum_{n=1}^{19} \cot^{-1}\left(1 + \sum_{p=1}^n 2p\right)\right)$ is : [JEE(Main)-2019]

(1) $\frac{22}{23}$ (2) $\frac{23}{22}$ (3) $\frac{21}{19}$ (4) $\frac{19}{21}$

IT0084

4. Considering only the principal values of inverse functions, the set $A = \left\{x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}\right\}$ [JEE(Main)-2019]

(1) is an empty set (2) Contains more than two elements
(3) Contains two elements (4) is a singleton

IT0086

5. If $\alpha = \cos^{-1}\left(\frac{3}{5}\right)$, $\beta = \tan^{-1}\left(\frac{1}{3}\right)$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\alpha - \beta$ is equal to : [JEE(Main)-2019]

(1) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (2) $\tan^{-1}\left(\frac{9}{14}\right)$ (3) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (4) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

IT0087

6. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, where $-1 \leq x \leq 1$, $-2 \leq y \leq 2$, $x \leq \frac{y}{2}$, then for all x, y, $4x^2 - 4xycos\alpha + y^2$

is equal to [JEE(Main)-2019]

(1) $4 \sin^2 \alpha - 2x^2y^2$ (2) $4 \cos^2 \alpha + 2x^2y^2$ (3) $4 \sin^2 \alpha$ (4) $2 \sin^2 \alpha$

IT0088

7. The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is equal to : [JEE(Main)-2019]

(1) $\pi - \sin^{-1}\left(\frac{63}{65}\right)$ (2) $\pi - \cos^{-1}\left(\frac{33}{65}\right)$ (3) $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$ (4) $\frac{\pi}{2} - \cos^{-1}\left(\frac{9}{65}\right)$

IT0089

8. If S is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(\frac{1}{21}\right) + \dots$, then $\tan(S)$ is equal to : [JEE(Main)-2020]

(1) $\frac{5}{11}$ (2) $-\frac{6}{5}$ (3) $\frac{10}{11}$ (4) $\frac{5}{6}$ IT0131

9. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$, Then the value of $\tan(M - m)$ is equal to: [JEE(Main)-2021]

(1) $2 + \sqrt{3}$ (2) $2 - \sqrt{3}$ (3) $3 + 2\sqrt{2}$ (4) $3 - 2\sqrt{2}$ IT0132

10. If the inverse trigonometric functions take principal values, then IT0133

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to : [JEE(Main)-2022]

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

EXERCISE (JA)

1. The value of $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^n2k\right)\right)$ is [JEE(Advanced) 2013, 2]

(A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

IT0092

2. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

List-II

P. $\left(\frac{1}{y^2}\left(\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right)^2+y^4\right)^{1/2}$ takes value 1. $\frac{1}{2}\sqrt{\frac{5}{3}}$

Q. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$, then possible value of $\cos\frac{x-y}{2}$ is 2. $\sqrt{2}$

R. If $\cos\left(\frac{\pi}{4}-x\right)\cos 2x + \sin x \sin 2x \sec x$ 3. $\frac{1}{2}$
 $= \cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4}+x\right)\cos 2x$,
 then possible value of $\sec x$ is

S. If $\cot(\sin^{-1}\sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$, $x \neq 0$, 4. 1
 then possible value of x is

Codes :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	4	2	1
(D)	3	4	1	2

[JEE(Advanced) 2013, 3, (-1)]

IT0093

3. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is [JEE(Advanced)-2014, 3]

IT0094

4. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option(s) is(are) [JEE(Advanced)-2015, 4]

(A) $\cos \beta > 0$ (B) $\sin \beta < 0$ (C) $\cos(\alpha + \beta) > 0$ (D) $\cos \alpha < 0$

IT0095

5. The number of real solutions of the equation

$$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right) \text{ lying in the interval } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ is } ______$$

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.)

[JEE(Advanced)-2018]

IT0096

6. Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.)

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e\left(\frac{x}{x-1}\right)$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$.

LIST-I

- P. The range of f is
Q. The range of g contains
R. The domain of f contains
S. The domain of g is

LIST-II

1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
2. $(0, 1)$
3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4. $(-\infty, 0) \cup (0, \infty)$
5. $\left(-\infty, \frac{e}{e-1}\right]$
6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$

The correct option is :

[JEE(Advanced)-2018]

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$ (B) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$
(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6$ (D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$

IT0097

7. The value of $\sec^{-1}\left(\frac{1}{4} \sum_{k=0}^{10} \sec\left(\frac{7\pi}{12} + \frac{k\pi}{2}\right) \sec\left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals

[JEE(Advanced)-2019, 3(0)]

IT0098

8. For any positive integer n , let $S_n : (0, \infty) \rightarrow \mathbb{R}$ be defined by $S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1+k(k+1)x^2}{x} \right)$, where for any $x \in \mathbb{R}$, $\cot^{-1}x \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) TRUE ?

[JEE(Advanced)-2021]

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{1+11x^2}{10x} \right)$, for all $x > 0$

IT0134

(B) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

9. Considering only the principal values of the inverse trigonometric functions, the value of

$\frac{3}{2} \cos^{-1} \sqrt{\frac{2}{2+\pi^2}} + \frac{1}{4} \sin^{-1} \frac{2\sqrt{2}\pi}{2+\pi^2} + \tan^{-1} \frac{\sqrt{2}}{\pi}$ is _____.

[JEE(Advanced)-2022]

IT0135

ANSWER KEY

Do yourself - 1

1. (i) C (ii) 1526 2. (i) $[-1, 1] - \{0\}$ (ii) $\left[-\frac{1}{3}, \frac{1}{2}\right]$ (iii) ϕ
 3. (i) $-1/3 \leq x \leq 1$ (ii) $\{1, -1\}$ 4. (i) $1 \leq x < 4$ (ii) $[-(1+\sqrt{2}), (\sqrt{2}-1)]$
 5. $x \in (-1/2, 1/2), x \neq 0$ 6. (i) $(3/2, 2]$ (ii) $(-2, 2) - \{-1, 0, 1\}$
 7. (i) $(-\infty, \ln \pi/2]$ (ii) $(0, \pi/2]$ 8. $[0, \pi]$ 9. A

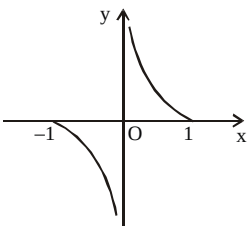
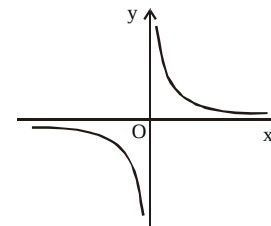
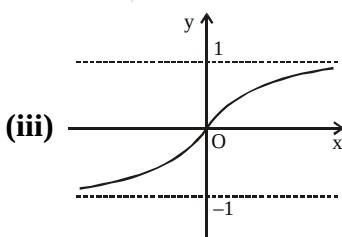
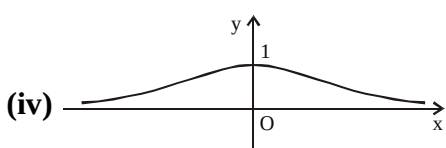
Do yourself - 2

1. (i) $\frac{15}{8}$ (ii) $\frac{4}{5}$ (iii) $\frac{4}{5}$ (iv) $2\sqrt{2}$ (v) $\frac{\sqrt{41}}{4}$ (vi) $\frac{63}{16}$
 2. (i) $\frac{1}{\sqrt{3}}$ (ii) $\frac{17}{6}$ (iii) $\frac{1}{2}$ (iv) -1 (v) 1 (vi) $\frac{1+3\sqrt{5}}{8}$ (vii) $\frac{6-4\sqrt{5}}{15}$
 3. (i) $\frac{5\pi}{6}$ (ii) $-\frac{\pi}{4}$ 4. (i) $\frac{1}{\sqrt{10}}$ (ii) $\frac{\pi}{6}$ (iii) $\frac{\sqrt{2}}{4}$
 5. (i) 2 (ii) $\frac{\sqrt{5}}{3}$ (iii) $\frac{1}{2\sqrt{2}}$

Do yourself - 3

1. (i) $\frac{\pi}{6}$ (ii) $\frac{\pi}{6}$ (iii) $\frac{\pi}{6}$ (iv) $-\frac{\pi}{6}$ (v) $-\frac{\pi}{3}$ (vi) $\frac{3\pi}{4}$ (vii) $\frac{\pi}{4}$
 5. $a \in [-2\pi, \pi] - \{0\}$ 6. C 7. B 8. A 9. D
 10. (i) $\pi - 4$ (ii) $4\pi - 10$ (iii) $2\pi - 6$ (iv) $4\pi - 10$ (v) $\frac{17\pi}{20}$ 11. $\frac{\pi}{2}$

Do yourself - 4

2. $\begin{cases} 1, & \text{if } a > 0 \\ -1, & \text{if } a < 0 \end{cases}$ 3. (i)  (ii) 
 (iii)  (iv)  all are identical.
 4. $k = \frac{11}{4}$ 5. A 6. (i) 0 (ii) 1

Do yourself - 5

3. (i) $\frac{\pi}{3}$ (ii) $\frac{1}{2}\tan^{-1}x$ 4. B 5. C

Do yourself - 6

3. (i) $2\tan^{-1}x - \pi$ (ii) $\pi - 2\sin^{-1}x$ (iii) $2\pi - 2\cos^{-1}x$
4. $\frac{1+xy}{x-y}$ 6. B

Do yourself - 7

1. (i) $\frac{1}{5}$ (ii) 12. (i) $(1, \infty)$ (ii) B 3. (i) $(-\infty, \sec 2) \cup [1, \infty)$
4. (i) $x = \frac{1}{2}\sqrt{\frac{3}{7}}$ (ii) $x = 3$ (iii) $x = 0, \frac{1}{2}, -\frac{1}{2}$ (iv) $\left[\frac{\sqrt{3}}{2}, 1\right]$ (v) $x = \frac{1}{2}, y = 1$
5. $x = 1; y = 2$ & $x = 2; y = 7$
6. (i) $(\cot 2, \infty) \cup (-\infty, \cot 3)$ (ii) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (iii) $\left(\frac{\sqrt{2}}{2}, 1\right) \cup \left(-1, -\frac{\sqrt{2}}{2}\right)$
7. $\left(\tan \frac{1}{2}, \cot 1\right]$ 8. C 9. $x \in (-1, 1)$
10. (i) $[-1, 0)$ (ii) $(-\infty \cot 3) \cup (\cot 2, \infty)$ (iii) $-\sin 1 < x \leq 1$
(iv) $\cos 2 < x \leq 1$ (v) no solution
11. (i) $\pm \frac{1}{\sqrt{3}}$ (ii) $x = 3$ (iii) $\pm \frac{1}{\sqrt{2}}$ (iv) $x = \frac{1}{2}$
12. C 13. A 14. D 15. C 16. B
17. (i) $x = \frac{4}{3}$ (ii) $x = \frac{a-b}{1+ab}$

Do yourself - 8

1. (i) $\tan^{-1}(x+n) - \tan^{-1}x$ (ii) $\frac{\pi}{4}$ (iii) $\frac{\pi}{2}$
2. (i) $\operatorname{arccot} \cot \left[\frac{2n+5}{n} \right]$ (ii) $\frac{\pi}{4}$
3. 54 4. 1

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	C	C	C	C	D	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	D	B	C	A	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	C	A	C	B	B	B	A	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	A	B	A,B	B,C	A,B,C,D	B,D	B,C	A,B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A,D	B	D	A,B	A,B,D	A,C	A	B	A,C,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	A	B	A,C,D	A,B,D	30.00	-1.00	C	A	
Q.10	A	B	C	D						
	p	p	p	s						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	1	2	0.00	2.50	2.50	3.14	12.50	20	10

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	3	4	1	3	3	4	4	C

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	B	3	B,C,D	2	A	0.00	A,B	2.35 or 2.36